

REVIEW

Neural substrates and neuroendocrine mechanisms underlying neophobia in vertebrates

Marquise S. Henry*, Michael B. Wilson and Christine R. Lattin

ABSTRACT

Neophobia, the avoidance of novel stimuli, is a widespread and ecologically relevant behavior observed across vertebrates and invertebrates. It is important for survival and adaptability, affecting how individuals navigate and use their environments. Neophobia might decrease an organism's exploitation of novel resources, but it also provides protection from potential dangers. Neophobia has been extensively studied in ecological and biomedical contexts, yet these two fields differ in their experimental paradigms, focal species and research goals. This Review uses a systematic approach to clarify the relationships between various 'neophobias' and the neural mechanisms that mediate them in vertebrates. Most vertebrate work focuses on mammals and, to a lesser extent, birds. In mammals and birds, both the hippocampus and amygdala (in birds, the medial ventral arcopallium) play a role in novelty evaluation and decision-making processes. In mammals, especially in the context of taste, food and odor neophobia, the insular and gustatory cortices also play a critical role. Serotonin, dopamine, glutamate, noradrenaline (norepinephrine) and glucocorticoids all exhibit context-, species- and life stage-dependent effects on neophobia. Fewer studies have examined immune or genetic mechanisms, and non-mammalian vertebrates are understudied relative to laboratory-bred rodents. In this Review, we also highlight persistent inconsistencies in how neophobia is defined and measured, emphasizing the need to distinguish it from neophilia and anxiety-related behaviors. Overall, this Review reveals both common mechanistic themes and major taxonomic and methodological gaps. We conclude by outlining key directions for future research, including broader comparative approaches and greater integration of ecological context into mechanistic studies of neophobia.

KEY WORDS: Neurobiology, Amygdala, Insular cortex, Serotonin, Dopamine, Glucocorticoids

Introduction: neophobia and why scientists study it

Neophobia, an aversive response to novelty, is a behavior observed in many animals (Martin et al., 2019; Mazza et al., 2021; Mettke-Hofmann et al., 2002), including both vertebrates and invertebrates. As environments shift from natural to human altered, organisms are increasingly exposed to novel stimuli they have previously not encountered. Willingness to interact with these novel elements might provide benefits, such as access to new foods or shelters (Greenberg and Mettke-Hofmann, 2001; Greggor et al., 2016), but it might also expose individuals to new risks (Sih et al., 2004). Theoretically, neophobia can be shown towards any kind of novelty,

but it is most often measured towards novel objects, environments, foods, odors or tastes; an individual might decide to approach or avoid a novel object, for example, or to feed on or reject a novel food item (Elvidge et al., 2016; Kimball and Lattin, 2024; Kiyokawa et al., 2024). Understanding the degree of neophobia of a particular species is an important piece of the puzzle in understanding its broader ecology. For example, ecologists often assess object, spatial and food neophobia to better understand traits such as landscape use, niche partitioning, and habitat and foraging specialization among different species (Greenberg, 1983, 1989; Many Birds Project et al., 2025). Neophobia is also of interest to ecologists because it affects the ability of different species to inhabit novel environments such as urban and suburban landscapes, to expand their range and to become invasive when introduced into new regions, phenomena that are increasingly important in a time of global anthropogenic change (Candler and Bernal, 2015; Krajcir et al., 2024; Magory Cohen et al., 2020; Moretti et al., 2015).

Neophobia is also an active area of biomedical research because of its implications for feeding behavior, neurodevelopment and understanding how the brain evaluates risk and reward. In particular, food neophobia plays an important role in human nutrition and health (for example, extreme avoidance of unfamiliar foods is associated with avoidance/restrictive food intake disorder, ARFID). Biomedical studies of neophobia often expose laboratory-bred rats and mice to novel foods and tastes to reveal the neural substrates and molecular mediators of responses to appetitive and aversive stimuli (Kesner et al., 1992; Ramírez-Lugo et al., 2015; Reilly, 2018; Yamamoto et al., 1995). Broader changes in novelty responses are sometimes linked to conditions including anxiety, autism spectrum disorder and other psychiatric disorders (Benito et al., 2018; Kashdan and Hofmann, 2008; Webb et al., 2010). Because of this connection to mental health, spatial and food neophobia are also sometimes examined in lab rodents as anxiety-linked behaviors (Bertin et al., 2018; Cole et al., 1988). There is also a wide literature examining humans' willingness to try novel foods, both as a potential measure of anxiety and with the hope of expanding the public's palate to increase consumption of different types of fish, insects and other foods (Sorokowska et al., 2025; Zickgraf and Elkins, 2018). This question has taken on particular importance as researchers and policymakers consider how to sustainably feed a growing global population in the face of climate change.

Across the different situations and species within which neophobia is investigated, it is remarkable that an individual's neophobia is often repeatable within the same context and novelty type (Bókony et al., 2012; Greenberg and Mettke-Hofmann, 2001), suggesting there are underlying differences in physiology or neurobiology that contribute to different responses among individuals. Further, variation in neophobia also exists across different species (Miller et al., 2022; Schaffer et al., 2021), but we do not fully understand the mechanisms driving these differences. This combination of inter- and intra-species variation in a behavior with clear outputs provides an excellent

Department of Biological Sciences, Louisiana State University, 202 Life Sciences Building, Baton Rouge, LA 70803, USA.

*Author for correspondence (mhennr55@lsu.edu)

© M.S.H., 0000-0002-6256-1423; M.B.W., 0009-0008-0073-7842; C.R.L., 0000-0003-4030-4212

Glossary

Approach–avoidance conflict

A situation in which an organism is both attracted to and repelled by the same stimulus (for example, appetizing food paired with a potentially aversive novel object).

Chemogenetic receptors hM3Dq and hM4Di

hM3Dq and hM4Di are two designer receptors exclusively activated by designer drugs (DREADDs), engineered G-protein-coupled receptors that no longer respond to their endogenous ligand, instead becoming activated by an otherwise inert synthetic compound such as deschloroclozapine. DREADDs are introduced into targeted neuronal populations using viral vectors, allowing researchers to selectively and reversibly manipulate neurons in awake behaving animals using systemic administration of the ligand. hM3Dq is excitatory, whereas hM4Di is inhibitory.

Cytochrome oxidase immunohistochemistry

Cytochrome oxidase is a mitochondrial enzyme required for adenosine triphosphate (ATP) production. More-active neurons need to generate more ATP, so cytochrome oxidase immunohistochemistry can map patterns of metabolic activity across different brain regions, informing researchers about the long-term energy demands of a neuron.

Gene module

A cluster of co-expressed genes identified using Weighted Gene Co-expression Network Analysis of transcriptome data. Co-regulated genes might have similar biological functions, such as sitting in the same biological pathway or contributing to the same biological process.

Immediate early genes

A class of genes that includes *c-fos*, *egr1* (*ZENK* in birds) and *Arc* that are transcribed quickly and transiently in response to cellular stimulation. These genes are often used as proxies for neuronal activity, and their mRNA or protein can be quantified, reflecting a time scale of minutes (mRNA) to hours (protein).

Neophilia

The preference for novel stimuli over familiar stimuli.

Neophobia attenuation

Decrease in an individual's neophobia towards a novel stimulus following repeated exposure.

Novelty indifference

Equal preference for novel and familiar stimuli.

Proteasome

A protein complex responsible for the degradation of damaged or misfolded proteins.

opportunity to examine how variation in neural and neuroendocrine physiology contributes to producing variation in behavior. Understanding the mechanisms behind neophobia could thus provide valuable insights into behavioral variation within and across species, as well as into anxiety-like behaviors. To aid this understanding, this Review aims to synthesize broad trends from vertebrate taxa across both biomedical and ecological studies related to (1) which parts of the brain and (2) which hormones, neurotransmitters or other molecular mediators are involved in neophobia. To achieve these aims, we performed a systematic review of the literature addressing the mechanisms underlying neophobia. We begin this Review by explaining which studies were included and which neophobia paradigms were considered. We then go on to discuss the relationship between various neural substrates and neuroendocrine mediators of neophobia in vertebrates. It is important to note that although we aimed to include a comprehensive review of this work across many vertebrate taxa, research in this field has primarily focused on rodents and, to a lesser extent, birds.

Let's talk about neophobia-testing paradigms

Our literature search strategy for this Review can be found in the [Supplementary Materials and Methods](#). We initially identified 503 articles that underwent further screening. To qualify for inclusion in

this Review, an article had to: (1) be a primary research article (not a review); (2) examine a physiological, neurobiological or endocrine mechanism; (3) focus on vertebrates (not invertebrates); and (4) test for neophobia [not neophilia (see [Glossary](#)) or an anxiety-linked behavior that might not indicate neophobia; a consideration discussed in more detail in the rest of this section]. In total, 157 articles were deemed relevant and included in this Review.

One issue we encountered when screening papers for inclusion is that many authors used the term 'neophobia' to describe behaviors that might in fact either be neophilia or a more generalized anxiety response ([Table 1](#)). To better understand how neophobia and neophilia differ, we can compare novel object tests and novel object recognition tests. Novel object tests involve placing an object that the test subject is unlikely to have previously encountered near a food source or nest containing offspring (Greggor et al., 2017; Kelly et al., 2022; Mettke-Hofmann et al., 2002; Perez et al., 2020). In this situation, individuals must decide between avoiding a potentially aversive novel object and approaching a motivating stimulus (their food or young). Animals that do not approach the motivating stimulus in this context are said to be neophobic.

A superficially similar assay, the novel object recognition test, examines the amount of time an individual animal spends with a novel object compared with a familiar object (Lueptow, 2017). In this experimental paradigm, there is no motivating factor near the novel object; any animals that choose to approach the novel object do so because they are inherently motivated to approach the object itself. Therefore, this assay measures an animal's neophilia, which some studies suggest might be a distinct behavior from neophobia (Mettke-Hofmann et al., 2002; Miranda et al., 2013). Because of this, we did not include studies using novel object recognition tests in this Review. For further information on the types of studies that were included or excluded, refer to [Table 1](#).

Based on the search terms and database we used, research on neophobia mechanisms has been published in 62 different journals ([Fig. S1A](#)). Most publications we reviewed investigated the roles of specific brain regions, specific neurotransmitters and their receptors, or other specific neuromodulators, such as beta amyloid plaques or primary cilia (65%; [Fig. S1B](#)). Roughly 12% of papers examined endocrine mechanisms, whereas 6% of papers looked at a combination of neural and hormonal mechanisms. Research on other types of physiological mechanisms, including immune responses and hypoxia, made up 17% of papers. Of the 159 accepted studies, 35 had an ecological focus rather than a biomedical one, and most of these studies used birds as their primary organism. Although the goal of this Review was to gather information on a wide variety of taxa, our search terms returned primarily rodent and bird studies. Below, we discuss the relevant findings in three separate sections summarizing trends in neural substrates, neuroendocrine mediators and other mechanisms involved in neophobia.

Neural substrates involved in neophobia

Research in both birds and rodents suggests that neophobia is mediated by coordinated activity in networks of brain regions involved in threat detection, sensory processing, learning, memory and decision making. Although mammalian and bird brains differ anatomically, structures such as the amygdala (mammals) and medial ventral arcopallium (birds), and the hippocampal formation in both lineages, appear to serve comparable functions in responding to and learning about novelty. A comparative approach can therefore help us distinguish conserved features of the neophobia response from species- or taxon-specific adaptations. We begin each section below by discussing the function and comparative anatomy of the brain

Table 1. Behavioral assays used in the literature for evaluating ‘neophobia’

Experimental assay	Description of assay	Variable(s) measured	Included in this Review?	Example reference(s)
Box escape test	The animal is placed into a plastic box with one opening. The plastic box is placed inside an open field arena.	Latency to exit the box	No	Rödíg et al., 2022
Corner test	The animal is placed into a novel cage, and its initial movements, especially toward corners, are tracked.	Time spent in corners; number of corner visits	No	Muntsant et al., 2019
Elevated plus maze	Subjects are placed into a raised plus-shaped (+) maze with four arms. Two of the arms are closed (with walls) and the other two are open (no walls).	Time spent in open versus closed arms; times entered open versus closed arms	No	Kjelstrup et al., 2002
Emergence test	Subjects are placed into a nest connected to a novel alley. Animals are then permitted to enter and exit the alley.	Number of emergences into the novel alley; time spent in the novel alley	No	Maren et al., 1993
Exploration test	Subjects are placed on a platform in a box. The platform has multiple holes in it. Interacting with the holes is recorded as a measure of exploration.	Locomotion; number of times subjects poke their nose into a hole	No	Arnold et al., 2015
Neophobia habituation (also called attenuation of neophobia)	The animal is fasted before being exposed to a type of novelty (e.g. food, object, taste) for multiple days.	Changes in latency to approach/feed from an initially novel stimulus over repeated exposures; amount consumed (for foods/tastes)	Yes	Ensminger and Westneat, 2012; Figueroa-Guzmán et al., 2006
Hyponeophagia test (also called novelty-suppressed feeding)	A fasted animal is placed into a novel environment containing a novel or familiar food.	Latency to approach and consume the food	Yes	Deacon, 2011; Greiner and Petrovich, 2020; McHugh et al., 2004; Shephard et al., 1982
Marble burying	The animal is placed in bedding with evenly spaced marbles, and its burying behavior is measured.	Number of marbles buried	No	Thomas et al., 2009
Novel food test	The animal is fasted before being provided with a novel food.	Latency to feed; amount consumed	Yes	Martin-Iverson et al., 1982; Royet and Pager, 1982
Novel object test	The animal is fasted before being provided with familiar food with a novel object on/in/ near it. Alternatively, novel objects can also be placed near entrances to nest sites containing young.	Latency to approach; latency to feed (when food is used as the motivating stimulus)	Yes	Greggor et al., 2017; Kelly et al., 2022; Lendvai et al., 2011; Perez et al., 2020
Novel object recognition	An animal is presented with a novel object and a familiar object.	Total time spent interacting with the novel object compared to time spent interacting with the familiar object	No	Lueptow, 2017
Novel taste or odor	An animal is water deprived before receiving access to a solution with a novel taste or odor.	Amount of the novel solution consumed (compared with a familiar solution or water on a different day)	Yes	Mason et al., 1978; Ramírez-Lugo et al., 2015; Rosenberg et al., 2016b

Because some of these assays do not require animals to approach or ingest something novel to obtain access to a motivating stimulus (e.g. food, water, offspring), they may measure animals’ neophilia, anxiety, compulsivity or exploration rather than neophobia. Studies using those assays were not included in this Review.

region in question, and then go on to consider evidence of their role in vertebrate neophobia.

Hippocampus

The hippocampus is involved in memory, spatial cognition, approach–avoidance conflict and regulation of the hypothalamic–pituitary–adrenal axis in birds and mammals, and likely in other amniotes as well (Broadbent et al., 2004; Cominski et al., 2014; Dedovic et al., 2009; Ito and Lee, 2016; Madison et al., 2023; Smulders, 2021). In rodents, the functions of the hippocampus are differentiated along the dorsoventral axis, with the dorsal hippocampus mediating spatial cognition and the ventral region more involved in decision making, anxiety, stress responses (Jimenez et al., 2018) and responding to aversive cues (Moser and Moser, 1998; Kjelstrup et al., 2002; Abela and Chudasama, 2013; Fanselow and Dong, 2010). Birds might show similar functional differentiation in the hippocampus, but along a rostral–caudal axis, where the rostral region is more involved in spatial cognition and the caudal region in responding to aversive

and stressful stimuli (Madison et al., 2023). Although reptiles, amphibians and fish all also have pallial regions of the brain that are functionally analogous to the hippocampus (Broglio et al., 2015; Rodríguez et al., 2002), our literature search did not return any papers on hippocampal regions and neophobia in these taxa.

Rodent research suggests that the role of the hippocampus in neophobia might depend on both the hippocampal subregion manipulated and the type of neophobia assessed. For example, ventral hippocampal lesions in rats decrease their latency to feed in a novel environment, suggesting that this subregion helps mediate spatial neophobia (Bannerman et al., 2002). In mice, a decrease in latency to feed on both familiar and novel foods in a novel environment is also seen following lesions to the entire hippocampus, supporting a role for the hippocampus in food and spatial neophobia (Deacon and Rawlins, 2005). However, inactivating the ventral hippocampus does not affect odor neophobia in rats (Shinohara and Yasoshima, 2021), suggesting that responses to different types of novelty might be mediated by different hippocampal subregions (Fig. 1).

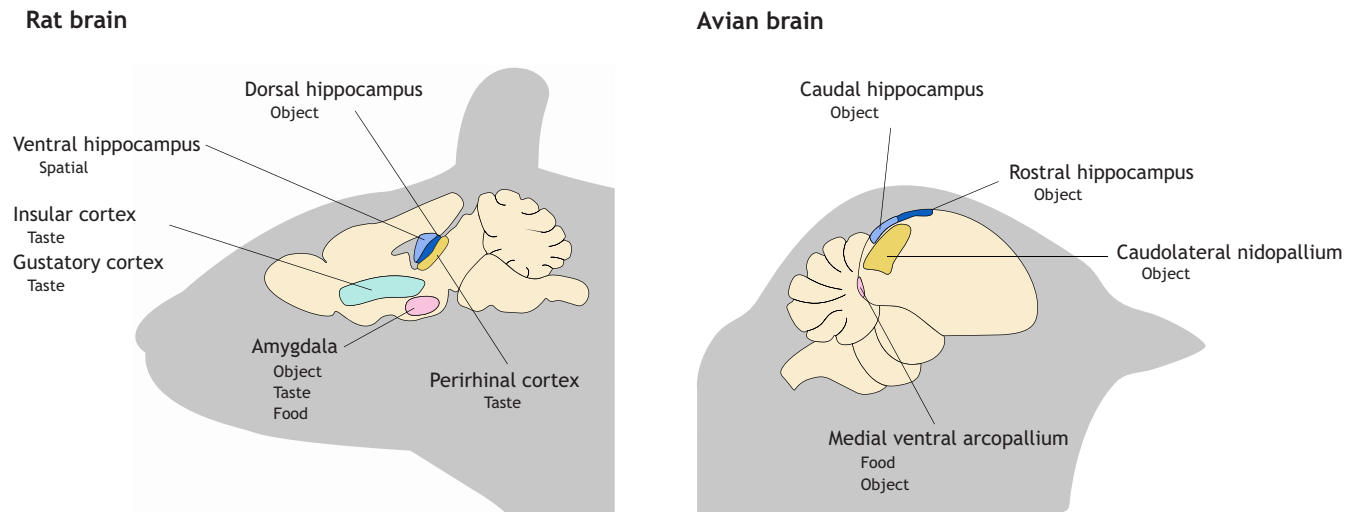


Fig. 1. Some of the brain regions involved in various types of neophobia in rats (left) and birds (right). Note that the bird medial ventral arcopallium was previously called the nucleus taeniae of the amygdala.

Further support for this idea comes from work showing that, unlike inactivation of the ventral hippocampus, lesions to the dorsal hippocampus of rats do decrease odor neophobia (Miller et al., 1986), but do not alter responses to novel tastes (Mitchell et al., 1993; Ramos, 2015) or environments (Mitchell et al., 1993). By contrast, lesioning the rat perirhinal cortex does decrease neophobia to palatable tastes (Ramos, 2015). The perirhinal cortex lies next to the hippocampus in the medial temporal lobe and connects to it mainly through the entorhinal cortex, a major gateway for cortical input into and out of the hippocampus (Agster and Burwell, 2009; Lavenex and Amaral, 2000). This suggests that hippocampus-adjacent brain regions in the rodent medial temporal lobe also contribute to behavioral responses to some types of novelty, and future studies should further examine brain regions connected to the hippocampus.

In female house sparrows (*Passer domesticus*), the caudal hippocampus shows large variation in gene expression (~12% of expressed genes showing differential expression) between neophobic and non-neophobic individuals (Lattin et al., 2022). Both male and female house sparrows also have increased c-Fos protein in the caudal but not rostral hippocampus following exposure to a novel object (Kimball et al., 2022). c-Fos is the product of an **immediate early gene**, which can be used to indicate recent neuronal activity. In the congeneric Eurasian tree sparrow (*Passer montanus*), neophobia correlates with the expression level of different **gene modules** in different parts of the hippocampus in male and female sparrows. Specifically, neophobia correlates with the expression of gene modules associated with steroid metabolism and neurotransmitters in the rostral hippocampus of male sparrows, and modules associated with learning, cytoplasmic translation and energy production in the caudal hippocampus of female sparrows, suggesting that neophobia might be linked to different subregions of the hippocampus and neural processes in male versus female songbirds (Lipshutz et al., 2026). Overall, these findings support a conserved role for the ventral (caudal in birds) hippocampus in at least some types of neophobia across both birds and mammals. Note, however, that some bird work suggests there could be sex differences in how the hippocampus contributes to neophobia that the rodent studies described above might have missed, given that most of those studies were done exclusively in male animals.

Amygdala

The mammalian amygdala is involved in decision making, memory, emotional processing (especially of fear, anxiety and aggression) and responding to novelty (Balderston et al., 2011; Sheth et al., 2008; Antoniadis and McDonald, 2001; Davis, 1992; Keifer et al., 2015; LeDoux, 2003). Nuclei in the amygdala are commonly divided into three groups: basolateral, cortical-like and centromedial (consisting of central and medial nuclei) (Sah et al., 2003). These nuclei differ in their projections, with the basolateral region projecting to various regions, including the hippocampus (Petrovich et al., 2001), and the medial and central regions projecting to the hypothalamus and bed nuclei of the stria terminalis (Reppucci and Petrovich, 2016; Sah et al., 2003). Although the basolateral and centromedial nuclei have been the focus of nearly all neophobia research in mammals, studies on mice have found that the cortical-like nuclei are required for unlearned aversive responses to predator odors and attraction to appetitive odors (Howe et al., 2026; Root et al., 2014), suggesting that the cortical amygdala may be a plausible candidate for a mediator of unlearned neophobia.

Amygdala-like regions have been identified across all major vertebrate groups. In birds, the arcopallium contains many nuclei whose homologies with mammalian amygdalar nuclei have been actively debated. The medial ventral arcopallium (formerly nucleus taeniae of the amygdala) shares connectivity, neurochemical and molecular features with the mammalian medial nucleus of the amygdala (Mello et al., 2019; Yamamoto et al., 2005), although recent molecular evidence suggests that bird arcopallium subdivisions correspond to mixtures of cortical and amygdalar cell populations instead of discrete mammalian homologs (Yu et al., 2023). Reptiles have an amygdaloid complex that includes cortical, basolateral (including the posterior dorsal ventricular ridge) and subpallial divisions (Lanuza et al., 1998; Martínez-García et al., 2002), while the amphibian amygdaloid complex contains vomeronasal (medial), olfactory/multimodal (lateral) and autonomic (central) subdivisions (Moreno and González, 2006). In teleost fish, the medial zone of the dorsal telencephalon is considered the best candidate for a pallial amygdala-like region, especially for functions associated with the mammalian basolateral nuclei including aversive learning, avoidance behavior and reward processing (Lau et al., 2011; von Trotha et al., 2014). Although our literature search only returned papers on the role of the amygdala in neophobia in mammals and birds, the existence of

these amygdala-like regions in other taxa creates opportunities for scientists to investigate their potential role in neophobia in reptiles, amphibians and fish.

Mammalian research on the role of the amygdala in neophobia often implicates the basolateral nuclei of the amygdala specifically in relation to novel food and taste avoidance in laboratory rodents. For example, lesions of the basolateral nuclei of the amygdala inhibit taste neophobia (Kesner et al., 1992; Kolakowska et al., 1985; Nachman and Ashe, 1974; Rollins et al., 2001; Shimai and Hoshishima, 1982), causing rats and mice to consume more of a novel solution. Similar effects are seen with inactivation of the amygdala through infusions of a GABA-A receptor agonist, muscimol (Shinohara and Yasoshima, 2019), which decreases taste neophobia in rats. Rats with larger basolateral nuclei of the amygdala also display greater object neophobia (Kiyokawa et al., 2017), suggesting that amygdala size might also influence responses to novelty.

Although the central and medial nuclei of the amygdala are also involved in taste neophobia in rodents, results relating to these nuclei are more complicated and methods dependent than for the basolateral nuclei of the amygdala. In rats, lesions to the medial nucleus decrease both taste and food neophobia (Kemble and Davies, 1981; Lin et al., 2009), whereas using the GABA agonists muscimol and baclofen to pharmacologically inhibit the medial nucleus has no effect (Arthurs et al., 2018). Excitation of rat medial nucleus neurons using the chemogenetic receptor **hM3Dq** also has no effect on the initial expression of taste neophobia, whereas chemogenetic inhibition using **hM4Di** enhances taste neophobia, the opposite result from that seen in lesion studies (Arthurs et al., 2018). It is possible that methodological issues (e.g. incomplete inhibition of neurons with GABA agonists, off-target effects of chemogenetic receptor ligands, lesion compensation) could cause these contradictory results, but there might also be as-yet uncharacterized cell-type or regional heterogeneity in the medial nucleus of the rodent amygdala that could explain these effects.

Studies using immediate early genes provide further support for the involvement of the amygdala in rodent and bird novelty responses. In rats, novel tastes increase c-Fos protein in both the basolateral and central nuclei of the amygdala (Lin et al., 2012; Wiaderkiewicz and Reilly, 2023), although an earlier study reported c-Fos elevation only in the central nucleus (Koh et al., 2003). In contrast, the medial nucleus of the amygdala does not show increased c-Fos protein in response to novel tastes, consistent with a more complicated role of this nucleus in neophobia (Lin et al., 2012; Wiaderkiewicz and Reilly, 2023). In rats, **cytochrome oxidase immunohistochemistry**, a longer-term measure of neuronal metabolic activity, reveals a different pattern, where metabolic activity of the basolateral nuclei of the amygdala is not elevated during an initial exposure to a novel vinegar solution, but is after a second exposure (Menchén-Márquez et al., 2023). These results suggest the basolateral nuclei of the amygdala contribute to both the initial detection of taste novelty in rodents and the evaluation and familiarization process that follows (Fig. 1). Increased c-Fos protein expression in the medial ventral arcopallium has also been observed in young chickens exposed to novel foods (Perez et al., 2020) and in adult house sparrows exposed to novel objects (Kimball et al., 2022) (Fig. 1), implying that the amygdala plays a conserved role in novelty assessment across mammals and birds.

Insular and gustatory cortices

The insular cortex is a mammalian brain region involved in affective processing (e.g. pain, fear and disgust) and risk-based decision making (Kober et al., 2008; Xue et al., 2010). A subsection of the insular cortex is the gustatory cortex, a region primarily responsible

for taste detection (Veldhuizen et al., 2007). In primates, the gustatory cortex includes parts of the anterior insula and frontal operculum (Chikazoe et al., 2019), whereas in rodents, the gustatory cortex is mostly located within the insular cortex (Samuelsen and Vincis, 2021). It is currently unclear whether birds, reptiles, amphibians or fish have pallial structures analogous to the insular or gustatory cortices (Biegler et al., 2021; Chen et al., 2013; Hegarty et al., 2024; Puelles, 2022), so all of the following studies were conducted on rats and mice. Because of their relatively poor eyesight and nocturnal/crepuscular lifestyles, rodents are probably more dependent on smell and taste to sense novelty than are visually oriented diurnal animals, such as many birds and lizards. Additionally, because lab rodents were bred from wild strains exposed to poisoning risks in human-associated environments, taste and smell neophobia are proposed to be particularly ecologically relevant traits in these species (Reidinger and Mason, 1983). In support of this idea, although lab-bred rodents are neophobic towards novel tastes, smells and foods, they tend to be neophilic towards novel objects (Kiyokawa et al., 2024; Koizumi et al., 2019). Note that much of this work on the mammalian insular and gustatory cortices, and their relationship to taste aversion learning, has been reviewed in detail elsewhere (Reilly, 2018).

In lesion studies, damaging the insular cortex reduces taste neophobia (Moraga-Amaro et al., 2014) (Fig. 1), regardless of whether the novel taste of saccharin is preferred to tap water (Stehberg et al., 2011). Insular damage also disrupts learning processes related to **neophobia attenuation** in conditioned taste aversion and taste-potentiated odor aversion assays (Lin et al., 2009; Roman and Reilly, 2007). Initial taste neophobia is also disrupted by disconnecting the insular and perirhinal cortices (Ramos, 2020), suggesting that these regions work together in processing information on taste novelty. Exposure to novel tastes increases c-Fos protein expression in the central insular cortex (Koh et al., 2003), and neuron excitability in the anterior insular cortex and prefrontal cortex (Kayyal et al., 2021), further supporting a role for the insular cortex in responding to novel gustatory stimuli in rodents.

Although the insular cortex is a critical hub for integrating sensory information (Zhang et al., 2024), there is good evidence that the reduced neophobia seen in the abovementioned lesion studies is not simply due to an animal's reduced ability to detect a novel odor or taste. When insular cortex-lesioned rats are exposed to a novel taste, a novel odor and a novel solution that acts as a trigeminal stimulant, lesions attenuate initial taste neophobia without affecting neophobic responses to the novel odor or trigeminal stimulant, demonstrating that insular cortex lesions do not affect sensory perception generally (Lin et al., 2009). Lin et al. (2009) also examined the attenuation of neophobia over time until consumption of a novel taste reached an asymptote, with the knowledge that rats typically plateau at a higher consumption of sweet and appetizing saccharin compared with water as they become accustomed to its novel taste. In that study, insular cortex-lesioned rats and sham-lesioned rats did not drink different amounts of saccharin at asymptote, showing that rats still perceived (and presumably preferred) the sweet flavor.

Ablations of the gustatory cortex attenuate neophobia towards bitter quinine, supporting the role of this insular subregion in responding to novel tastes in rats (Kiefer et al., 1982) (Fig. 1). Disrupting communication between the rat basolateral nuclei of the amygdala and gustatory cortex using a cross-disconnection approach (lesioning the basolateral nuclei in one hemisphere and the gustatory cortex in the other) also attenuates neophobia, whereas disrupting communication between the medial nucleus and gustatory cortex has no effect (Lin and Reilly, 2012). Further, disrupting communication between the medial and basolateral nuclei of the amygdala also does

not affect neophobia. Collectively, these studies demonstrate that, in rats, communication between the basolateral nuclei of the amygdala and gustatory cortex is necessary for either novel taste recognition or the aversive behavioral response that follows, and that the medial nucleus can also affect neophobia through a different pathway independent of both the basolateral nuclei of the amygdala and gustatory cortex (Fig. 2).

Time-dependent patterns of protein synthesis and degradation in the gustatory cortex appear to be important for rodents to learn about new tastes. Shortly after exposure to a novel taste, rats show a transient reduction in activity of the **proteasome** and there are fewer proteins tagged for degradation in the gustatory cortex (Rosenberg et al., 2016b). This early period of protein stabilization is followed by an increase in proteasome activity ~4 h after novel taste exposure, a process that is dependent on activation of the glutamatergic NMDA receptor (Rosenberg et al., 2016a). These results suggest that initial exposure to a novel taste temporarily induces protein stabilization in the rat gustatory cortex, possibly to help encode that novel experience, followed by NMDA receptor-dependent protein degradation, which might result in the synaptic plasticity and remodeling needed for memory consolidation and the attenuation of taste neophobia.

Neuroendocrine mediators of neophobia

Research on neophobia mechanisms identified by our literature search has examined not just what brain regions are involved in responding to novelty but also the hormones and neurotransmitters that are linked to neophobia, whether through local action within the brain or systemic effects. Both bird and rodent studies have identified multiple neuroendocrine pathways that impact novelty-related

behaviors. Most neurotransmitter work has been done in lab-bred rodents, consistent with these model organisms being used more for mental health research. In contrast, the effects of glucocorticoid hormones on neophobia have been particularly well studied in avian systems. This is because glucocorticoids are affected by artificial selection when animals are bred in captivity (Künzl and Sachser, 1999; Løtvedt et al., 2017; Trut et al., 2009); scientists interested in the full range of glucocorticoid responses to potential stressors (which might include novel stimuli) therefore often prefer to use wild animal models such as songbirds. Below, we discuss findings on the role of glutamate, serotonin, dopamine, noradrenaline (norepinephrine) and glucocorticoids in neophobia.

Glutamate

Glutamate is an excitatory neurotransmitter with two main receptor subtypes: ionotropic, ligand-gated ion channels that include AMPA, NMDA, delta and kainate receptors, and metabotropic, G-protein-coupled receptors that can be either inhibitory or activating (Reiner and Levitz, 2018). Although both ionotropic and metabotropic glutamate receptors are conserved across vertebrates (Moroz et al., 2021; Ramos-Vicente et al., 2018), the glutamate literature on neophobia is dominated by rat studies of the insular cortex. As mentioned above, glutamatergic NMDA receptors are involved in the proteasome response of the gustatory cortex to novel tastes in rats (Rosenberg et al., 2016a). Infusion of glutamate into the insular cortex impairs attenuation of taste neophobia, while antagonism of the excitatory metabotropic glutamate receptor 1 facilitates this attenuation (Ramírez-Lugo et al., 2015).

The role of NMDA receptors specifically in the attenuation of neophobia is more complicated. Microinfusion of a non-competitive NMDA receptor antagonist, MK-801, into the insular cortex of rats prevents the attenuation of taste neophobia without affecting the magnitude of the initial neophobic response or the asymptotic level of intake once infusions stop (Figueroa-Guzmán et al., 2006). In contrast, a competitive NMDA receptor antagonist, AP-5, infused into the insular cortex has no effect on initial neophobia or its habituation (Gutiérrez et al., 2003). There are several possible explanations for these different effects of competitive and non-competitive NMDA antagonists. Figueroa-Guzmán et al. (2006) suggest the concentration of competitive antagonist used by Gutiérrez et al. (2003) might have been too low to fully block NMDA receptors. However, this dose blocked conditioned taste aversion learning in the same animals, suggesting that it did affect NMDA receptors. A second possibility relates to differences in the action of non-competitive versus competitive antagonists. Non-competitive NMDA antagonists such as MK-801 are open-channel blockers, requiring the channel to be activated by glutamate binding and depolarization before the drug can act; thus, they preferentially block the most actively signaling receptors. Competitive antagonists such as AP-5 bind regardless of channel activation state and block receptors more uniformly. If neophobia attenuation depends on a specific subpopulation of insular cortex NMDA receptors engaged during repeated taste exposures, non-competitive blockade might disrupt this process more than competitive blockade. Finally, MK-801 has been shown to inhibit recognition memory (de Lima et al., 2005; Hoffman and Basurto, 2013; Oliveira et al., 2021) and reduce anxiety-like behaviors in mice (Nilsson et al., 2007), raising the possibility of off-target effects. However, the behavioral specificity of the Figueroa-Guzmán et al. (2006) result (no effect on initial neophobia or asymptotic intake, only on the attenuation of neophobia) does not support a generalized anxiolytic explanation.

Glutamate signaling in the insular cortex is also modulated by intracellular kinases. Protein kinase C can enhance NMDA

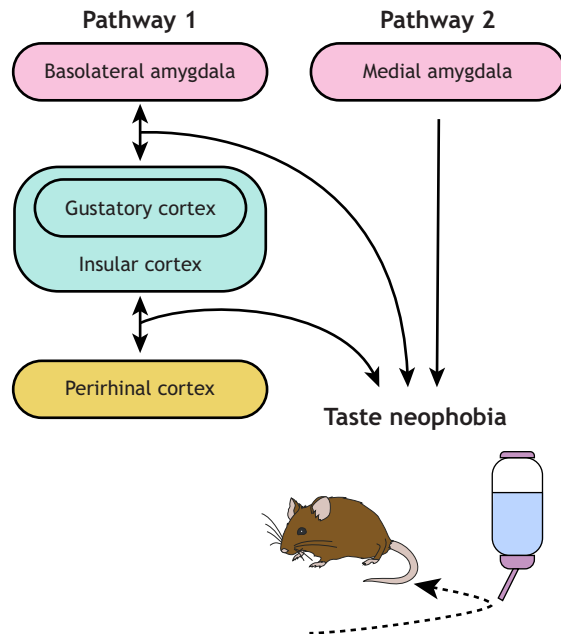


Fig. 2. Different neural circuits mediating taste neophobia in rodents. Inactivating either part of the reciprocal connection between the basolateral nuclei of the amygdala ('basolateral amygdala') and the gustatory cortex or that between the insular and perirhinal cortices attenuates taste neophobia. Lesioning the medial nucleus of the amygdala ('medial amygdala') also decreases taste neophobia. However, disconnecting the medial nucleus of the amygdala from either the basolateral amygdala or the gustatory cortex does not affect taste neophobia, indicating that the basolateral amygdala–gustatory cortex and the medial nucleus of the amygdala mediate taste neophobia through separate pathways.

function (and thus excitatory glutamatergic signaling) via an indirect pathway involving Src family tyrosine kinases (Lu et al., 1999). However, blocking protein kinase C in the rat insular cortex does not affect taste neophobia attenuation across multiple exposures (Núñez-Jaramillo et al., 2007), suggesting that protein kinase C-mediated signaling is not required for safe taste learning. Collectively, these results demonstrate that glutamatergic signaling in the rat insular cortex affects responses to novel tastes, but its influence depends on which receptor subtype is involved and, perhaps, the pattern of activation. Whether the same mechanisms apply across other vertebrates is unknown, as work thus far has been conducted almost exclusively in rats and mice.

Serotonin

Serotonin is a neurotransmitter produced in the raphe nuclei of the brainstem in vertebrates (birds, mammals, reptiles, fish and amphibians) (Branchereau et al., 2000; Fujita et al., 2023; Lillesaar, 2011; Steinbusch, 1981; Wolters et al., 1985) that influences memory, mood and stress responses (Berger et al., 2009). Even though the serotonergic system is highly conserved across vertebrates, similarity in the locations and projection patterns of serotonin neurons across vertebrates does not necessarily mean their functions are conserved, especially given the complexity of serotonergic effects on neophobia described below.

Serotonergic signaling is often implicated in mediating responses to novelty, but the direction of its effects differs between correlational and causal studies. In a novel object paradigm where rats had to approach a novel object to access their preferred cage area, the number of serotonergic neurons and their c-Fos protein expression in the dorsal raphe increased linearly across neophobic, novelty **indifferent** and neophilic rats (Kiyokawa et al., 2024). This finding suggests that neophobia is associated with lower serotonergic signaling. However, in a different study, mice lacking tryptophan hydroxylase 2, the rate-limiting neural enzyme for serotonin synthesis, were faster to feed in a novel environment (Mosienko et al., 2012), suggesting that the loss of serotonin can also reduce neophobia. One way to reconcile these apparently contradictory results is by distinguishing between tonic (sustained, baseline) and phasic (brief, stimulus-evoked) serotonergic signaling, and to consider the action of different serotonin receptor subtypes, which can have opposing behavioral effects. For example, transient activation of dorsal raphe serotonergic neurons in mice suppresses locomotion, whereas repeated daily activation of these neurons over 3 weeks increases locomotion (Correia et al., 2017). Similarly, loss of 5-HT_{1A} receptors produces an anxiety-like phenotype (Parks et al., 1998; Ramboz et al., 1998), and central overexpression of 5-HT_{1A} receptors reduces anxiety-like behaviors (Kusserow et al., 2004), whereas 5-HT_{2C} receptors can act in the opposite direction, with forebrain overexpression increasing anxiety-like behavior in mice (Kimura et al., 2009), consistent with the anxiogenic effects of 5-HT_{2C} agonists in brain regions such as the ventral hippocampus (Alves et al., 2004). Although they did not use neophobia behavioral paradigms, the abovementioned findings can help us understand how two studies that examined the number of serotonergic neurons and a serotonin synthesis knockout without considering timing differences or different receptor subtypes could indicate seemingly opposite relationships between serotonin and neophobia that might both be valid.

In wild birds, genetically linked variation in serotonin transporter and receptor genes has been correlated with individual differences in neophobia. For example, in great tits (*Parus major*), some polymorphisms in the gene for the serotonin transporter (SERT), which is responsible for serotonin reuptake from the synaptic cleft,

are associated with increased latency to approach a novel object (Timm et al., 2018), whereas single nucleotide polymorphisms in the SERT promoter in the same species are associated with the opposite pattern, shorter latency to approach a novel object (Riyahi et al., 2022). Different polymorphisms can increase or decrease the activity of a transporter (Hahn and Blakely, 2007), so these findings are not necessarily contradictory. In female house sparrows, expression of serotonin receptor 5A, one of the least-well understood serotonin receptors (Goodfellow et al., 2012), is also higher in the caudolateral nidopallium of less neophobic birds compared with more neophobic birds (Lattin et al., 2022).

Overall, the published literature shows that serotonergic signaling influences neophobia in both mammals and birds, albeit in complex ways. Given the strong conservation of this signaling system across vertebrates, serotonin is likely to modulate neophobia in reptiles, fish and amphibians as well.

Catecholamines

Dopamine and noradrenaline are catecholamine neurotransmitters implicated in how animals respond to novelty. Dopamine is produced in midbrain dopaminergic cell groups corresponding to the substantia nigra and ventral tegmental area in mammals and functionally similar regions in other vertebrates, as well as in the hypothalamus (Durstewitz et al., 1999; Juárez Olguín et al., 2016; Reiner et al., 1998), and is involved in reward (Luo and Huang, 2016) and decision making (Friston et al., 2014). Noradrenaline, by contrast, is synthesized primarily by noradrenergic neurons in the locus coeruleus of the pons, which projects to the hippocampus, amygdala and neocortex (Berridge and Waterhouse, 2003; McCall et al., 2017). Noradrenaline mediates arousal and vigilance, and is highly conserved across vertebrates (Hao et al., 2025; Romero and Gormally, 2019). Both neurotransmitters have been linked to neophobic behavior, though, as described below, they act through distinct, and in some cases contradictory, mechanisms.

Dopaminergic systems often seem to facilitate exploratory behavior and attenuate neophobic responses (Peters et al., 2007; Pogorelov et al., 2005). In rodents, treatment with a non-selective dopamine agonist, apomorphine, decreases taste neophobia (Kesner et al., 1975), and c-Fos protein in dopaminergic (tyrosine hydroxylase positive) neurons in the ventral tegmental area is higher in rats that are neophilic towards novel objects than in indifferent rats with no preference for novel or familiar stimuli (Kiyokawa et al., 2024). Furthermore, more-neophobic female house sparrows show reduced hippocampal expression of dopamine receptors 1, 2 and 5 compared with less-neophobic females (Lattin et al., 2022). Variation in the dopamine receptor D4 gene (*DRD4*) has also been linked to novelty responses in wild collared flycatchers (*Ficedula albicollis*), where one polymorphism in *DRD4* is associated with increased risk-taking behavior, and another shows a weaker association with neophobia (Garamszegi et al., 2014).

However, not all dopaminergic manipulations affect neophobia. For example, in mice, injection of hippocampal cerebral dopamine neurotrophic factor, a protein that supports the survival of dopamine neurons (Lindholm and Saarma, 2022), does not affect object neophobia (Kempainen et al., 2015). However, this is a more indirect way of manipulating the dopamine system, and dopamine neuron protection might not be necessary for the normal expression of neophobia.

Noradrenaline also plays a role in neophobia in rodents. Reducing noradrenaline in the forebrain using 6-hydroxydopamine, a neurotoxin that causes cell death through oxidative stress and mitochondrial dysfunction (Glinka et al., 1997), decreases initial

taste neophobia in rats (Mason et al., 1978; Mason and Fibiger, 1979). However, 6-hydroxydopamine can also cause the cell death of dopaminergic neurons (Smith et al., 2002), which also could have affected the behavior observed in these studies. More recent work suggests that noradrenergic signaling in the nucleus accumbens of rats is also required for corticosterone-mediated learning about novel tastes (Wichmann et al., 2012). Blocking noradrenergic activity in the nucleus accumbens shell, but not core, prevents the increased habituation to novel tastes normally induced with a glucocorticoid agonist, suggesting that glucocorticoid-mediated neophobia is partially dependent on noradrenaline.

Interestingly, the direction of noradrenaline effects on initial neophobia can also vary across individuals. The work described above was done in an unselected population of CD-COBS rats, a Sprague–Dawley-derived outbred strain that is generally less active and exploratory than Wistar rats (Walker et al., 2009). In a normal population of Wistar rats, ~80–90% can be classified as high or low responders in an open field test, an assay which assesses locomotion and anxiety-like behavior by measuring the time an animal spends in the center versus periphery of an open arena. High responders generally show greater exploration and locomotion, whereas low responders move and explore much less. This high/low responder behavioral phenotype also covaries with the noradrenaline state of the nucleus accumbens during an environmental challenge (Cools et al., 1990, 1993). Beta-adrenergic receptor manipulation of the basolateral nuclei of the amygdala produces opposite effects on initial food neophobia in the two groups, where a beta-receptor antagonist, propranolol, reduces neophobia in the high responder group but has no effect in the low responder group, whereas a beta-receptor agonist reduces neophobia in low but not high responders (Roozendaal and Cools, 1994). This research demonstrates that there may be two different neurobiological ‘paths’ to food neophobia in Wistar rats, one that relies upon high beta-adrenergic receptor activity in the basolateral amygdala (in high responders) and the other on low beta-adrenergic receptor activity in the basolateral amygdala (in low responders).

The effects of noradrenaline on neophobia can also depend on novelty type, and a single manipulation can push different types of neophobia in opposite directions. Cole et al. (1988) demonstrate this within the same animals: lesioning the dorsal noradrenergic bundle, which supplies noradrenaline to the forebrain, simultaneously increases rats’ neophobia towards a novel environment and reduces neophobia towards a novel food. Lesioned rats are slower than controls to contact familiar food in a novel open field, but show reduced avoidance of a novel food, eating it about as readily as

familiar food (Cole et al., 1988). The novel food effect agrees with the taste results noted above (noradrenaline depletion decreasing taste neophobia; Mason et al., 1978; Mason and Fibiger, 1979), and the novel environment result is similar to work in dopamine β -hydroxylase knockout mice, which cannot synthesize noradrenaline and also reduce their feeding on familiar food in novel environments (Lustberg et al., 2020). Further work showing that different types of novelty may be mediated differently comes from a behavioral study in house sparrows, where spatial and object neophobia were found to be uncorrelated within individuals (Dusang et al., 2025), supporting the hypothesis that different types of neophobia can have distinct neuroendocrine mechanisms.

However, there are two potential complications to the rodent work. First of all, dorsal noradrenergic bundle lesions mostly deplete cortical noradrenaline, but they also lower noradrenaline in the hypothalamus, and Cole et al. (1988) attribute the different effects on spatial and food neophobia to noradrenaline loss in different regions instead of a single mechanism affecting different types of neophobia differently. Second, dopamine β -hydroxylase knockout mice can also show increased dopamine in some brain regions (Bourd  lat-Parks et al., 2005), so their behavior might partly reflect altered dopamine signaling rather than lost noradrenaline signaling. Together, these studies suggest that dopamine and noradrenaline both affect neophobic responses, but through distinct mechanisms. Dopamine tends to facilitate exploration and reduce neophobia, whereas noradrenergic signaling effects depend on novelty type (Fig. 3). This suggests that neophobia is not mediated by a single catecholamine pathway but instead by the interplay of at least two distinct systems.

Glucocorticoids

Glucocorticoids are steroid hormones released from the hypothalamic–pituitary–adrenal axis (the hypothalamic–pituitary–interrenal axis in fish) that help regulate homeostasis, and which increase in response to environmental stressors (Sapolsky et al., 2000). In rodents, birds, reptiles and most amphibians, the primary glucocorticoid is corticosterone; in other mammals and fish, it is cortisol (Narayan et al., 2013; Sadoul and Geffroy, 2019; Sopinka et al., 2015). In mammals and birds, glucocorticoid effects on the brain occur through their binding to two different populations of receptors: a higher-affinity mineralocorticoid receptor, thought to mediate hormone effects at basal or baseline concentrations, and a lower-affinity glucocorticoid receptor, which shows substantial binding only at stress-induced or acute concentrations (Breuner and Orchinik, 2009; Reul et al., 1987). An *in situ* hybridization study in European starlings (*Sturnus vulgaris*) found glucocorticoid receptor

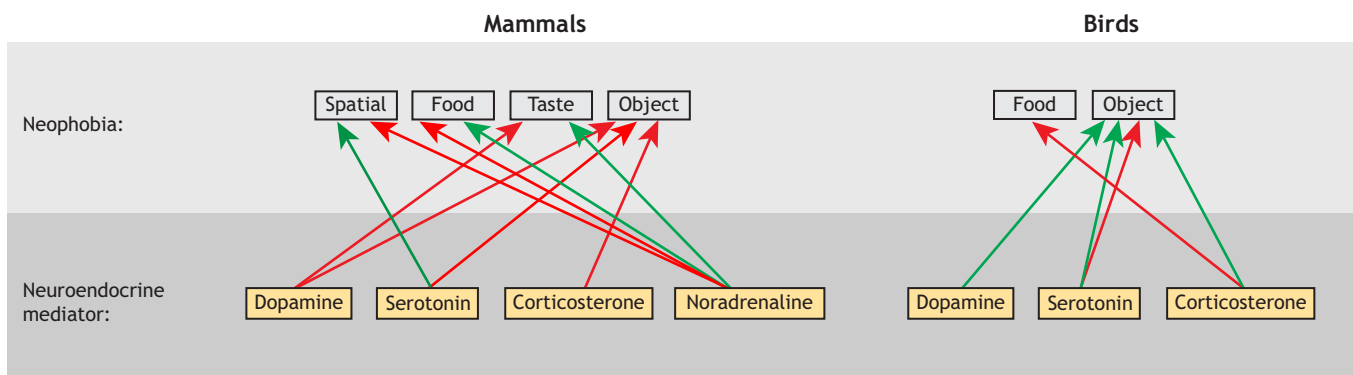


Fig. 3. An overview of the currently known relationships between different neurotransmitters and hormones, and the types of neophobia they affect, in mammals (mostly rodents) and birds. Red arrows represent a negative correlation. Green arrows represent a positive correlation.

mRNA throughout the brain, whereas mineralocorticoid receptor mRNA was mostly observed in the hippocampus, hypothalamus and optic tectum (Dickens et al., 2009; see also Senft et al., 2016). Broadly similar patterns in receptor distribution occur in mammals, with widespread brain expression of glucocorticoid receptors, and mineralocorticoid receptor expression restricted to the hippocampus, although the finer details of receptor distribution vary by species (Seckl et al., 1991; Van Eekelen et al., 1988). As mentioned above, most work on glucocorticoids and neophobia has been conducted in birds.

Relationships between glucocorticoids and neophobia are complex, to say the least (Table 2). Many animals show context-, sex- and species-specific associations between glucocorticoids and neophobia; there are also often differences among studies that might be related to different effects of baseline versus stress-induced glucocorticoids on behavior, whether glucocorticoids were manipulated (e.g. by injection with corticosterone or at the level of the receptor via glucocorticoid receptor antagonists), and when glucocorticoid concentrations were assessed relative to

neophobia. Injection of mitotane, a drug that decreases baseline and stress-induced corticosterone concentrations, also decreases object neophobia in house sparrows (Kelly et al., 2022). However, inhibiting glucocorticoid receptors with the antagonist mifepristone does not affect neophobia in the same species (Henry et al., 2025), suggesting that baseline corticosterone acting through mineralocorticoid receptors might be more important in neophobia than stress-induced corticosterone binding to glucocorticoid receptors. Consistent with this hypothesis, there is a trend for lambs with higher baseline cortisol to feed less from novel feeders (Rice et al., 2016). However, chickadees with higher baseline corticosterone are less, not more, neophobic towards novel foods (Prasher et al., 2019), which does not fit this pattern.

Effects of glucocorticoids on neophobia might also be life-stage dependent, as administering postnatal corticosterone at 8-, 9- and 11-days post-hatch does not affect object neophobia in juvenile or adult house sparrows (Kimmitt et al., 2025). However, receiving exogenous injections of corticosterone as nestlings decreases object neophobia several weeks later in male, but not female, zebra finches

Table 2. An overview of studies investigating the relationship between glucocorticoid hormones and neophobia

Source	Animal species or strain	Timing of sample collection, if any	Endogenous or exogenous hormone?	Type of measure	Baseline or stress induced?	Type of neophobia	Main findings
Apfelbeck and Raess, 2008	European starlings (<i>Sturnus vulgaris</i>)	2 h after novel object testing	Endogenous	Plasma	Baseline*	Object	Corticosterone increased following novel object trials
Arnold et al., 2016	Blue tit (<i>Cyanistes caeruleus</i>)	24 h after capture and shortly after behavioral testing	Endogenous	Plasma	Both	Spatial	No relationship between glucocorticoids and neophobia
de Bruijn and Romero, 2020	European starlings (<i>Sturnus vulgaris</i>)	No samples were collected	Exogenous and endogenous	None	Neither	Object	Exposure to a restraint stressor increased neophobia, but injection of corticosterone did not
Garamszegi et al., 2012	Collared flycatchers (<i>Ficedula albicollis</i>)	2–3 h after behavioral testing	Endogenous	Fecal glucocorticoid metabolites	Neither	Object	No relationship between glucocorticoids and neophobia
Kelly et al., 2022	House sparrow (<i>Passer domesticus</i>)	1–2 days after novel object trials	Endogenous	Plasma	Stress-induced	Object	No relationship between natural variation in stress-induced corticosterone and neophobia, but blocking corticosterone synthesis reduced neophobia
Kimmitt et al., 2025	House sparrow (<i>Passer domesticus</i>)	No samples were collected	Exogenous	None	Neither	Object and spatial	Treatment with corticosterone as juveniles did not affect object or spatial neophobia as adults
Prasher et al., 2019	Black-capped chickadees (<i>Parus atricapillus</i>)	The last day of capture	Endogenous	Plasma	Baseline	Food	More neophobic individuals had lower baseline corticosterone
Rice et al., 2016	Merino lambs	Between days 4 and 6	Endogenous	Plasma	Baseline	Food and object	More neophobic individuals tended to have higher cortisol, but this was not significant ($P=0.10$)
Spencer and Verhulst, 2007	Zebra finch (<i>Taeniopygia guttata</i>)	Individuals received corticosterone injections as nestlings. Corticosterone was assessed at 12 days old, 2 weeks before trials.	Exogenous	Plasma	Both	Object	Injection with corticosterone as nestlings decreased object neophobia weeks later in males but not females

Included are when and how glucocorticoids were measured, and the main findings. *Blood was collected 2 h after testing, but the authors acknowledge this might not have been enough time for circulating corticosterone to return to baseline.

(Spencer and Verhulst, 2007). Glucocorticoid signaling can also play a role in the attenuation, as well as the initial expression, of neophobia. For example, agonist-mediated activation of the glucocorticoid receptor (using the selective glucocorticoid agonist RU-28362) in the nucleus accumbens shell of rats facilitates their habituation to novel tastes (Wichmann et al., 2012).

Overall, the literature suggests that glucocorticoids can affect both the initial expression and attenuation of neophobia, but the effects may be context, sex, species and developmental stage dependent (Fig. 3). Furthermore, the relationship between glucocorticoids and neophobia is at least somewhat bidirectional: following exposure to a novel object, European starlings show an increase in baseline corticosterone levels (Apfelbeck and Raess, 2008). One limitation of the literature is that most studies on glucocorticoids and neophobia have not manipulated hormones or receptors; further experimental work would help reveal the mechanistic role of glucocorticoid hormones in neophobia.

Other mechanisms associated with neophobia

A small number of published studies have investigated other potential physiological or neurobiological mechanisms underlying neophobia in vertebrates (Fig. S1). For example, there is some evidence that immune activation can increase neophobia. In mice, stimulation of T-cells with either staphylococcal enterotoxin A (Kawashima and Kusnecov, 2002) or staphylococcal enterotoxin B (Kusnecov et al., 1999) increases taste neophobia in novel environments. Neophobia might be partly affected by downstream activation of the HPA axis and glucocorticoid release caused by cytokine production from activated T-cells (Silverman et al., 2005). Alternatively, a neophobic response to novel tastes in these studies might be a result of the Garcia effect, a learned avoidance of novel foods and tastes when previous consumption is associated with sickness (Garcia et al., 1966; Garcia and Koelling, 1966; Rivi et al., 2024). Although more work is needed to clarify the mechanisms underlying these effects, these studies suggest that immune system activation can increase the avoidance of novelty, potentially reflecting an adaptive behavioral strategy to reduce exposure to additional dangers while animals are ill.

Rodent research suggests additional pathways that can influence neophobia in specific contexts. For example, transgenic mice with reduced levels of beta-endorphin display increased spatial neophobia following stress in the form of forced swimming (Barfield et al., 2013). Beta-endorphins can modulate dopamine release, primarily by acting via mu-opioid receptors in brain regions such as the ventral tegmental area (Johnson and North, 1992), implying that beta-endorphins could affect neophobia through monoaminergic pathways as well.

Exposure to extremely low doses of the environmental pollutant 2,3,7,8-tetrachlorodibenzo-*p*-dioxin (TCDD) rapidly induces strong avoidance of a novel food in both rats and mice (Lensu et al., 2011). TCDD alters gene expression by binding to the aryl hydrocarbon receptor, a transcription factor activated by ligands from environmental pollutants (Couvreur et al., 2026) as well as plant metabolites (Bjeldanes et al., 1991; Denison and Nagy, 2003). These findings suggest that neophobia can be impacted by an individual's diet and exposure to chemical pollutants. Overall, these studies reveal that a diverse set of biological systems can influence neophobia.

Initial neophobia versus neophobia attenuation

Many of the studies described above examined both an individual's initial response to a novel stimulus and the attenuation of neophobia upon repeated presentations. Avoidance is a memory-dependent process, and it typically declines with repeated exposure unless a

novel stimulus is paired with something aversive. Several lines of evidence demonstrate that the initial expression of neophobia and its attenuation over repeated exposures do not always depend on the same neural substrates or neurotransmitter systems, and an experimental manipulation can affect one without affecting the other.

Some of the clearest evidence of dissociations between neural substrates involved in neophobia versus its attenuation comes from work in the rat amygdala and hippocampus. For example, inactivating the basolateral nuclei of the amygdala with the GABA agonist muscimol reduces initial taste neophobia, but does not affect the attenuation of neophobia (Shinohara and Yasoshima, 2019). In the ventral hippocampus, however, inactivation does not affect initial odor or taste neophobia but causes greater attenuation of neophobia towards novel odors during a second exposure (Shinohara and Yasoshima, 2021). This effect is specific to novel odors, and does not affect tastes, also demonstrating that different brain regions might mediate responses to different types of novelty that rely on different sensory modalities.

The neurotransmitter literature shows a similar pattern, where glutamatergic signaling via NMDA receptors is more involved in the attenuation of taste neophobia than in initial neophobia. When insular cortex or basolateral amygdala NMDA receptors are blocked with an antagonist in rats, neophobia attenuation is prevented but initial taste neophobia is left unaffected (Figueroa-Guzmán et al., 2006, 2008). Noradrenergic signaling in the basolateral amygdala, by contrast, appears to act mostly on the initial neophobic response. Bilateral infusion of noradrenaline into the basolateral nuclei of the amygdala shifts rats' food preferences towards familiar chow and away from novel foods (Borsini and Rolls, 1984). Serotonergic manipulation in the basolateral amygdala has no effect on food preferences (Borsini and Rolls, 1984), demonstrating that the effect is neurotransmitter specific and not a general consequence of disrupting amygdala signaling.

One final consideration is that even though initial neophobia and its attenuation can be mediated by distinct neural substrates and neurotransmitter systems, both can still co-vary at the level of the individual. For example, in both house sparrows and Eurasian tree sparrows, individual birds that are faster to eat novel foods and familiar foods in the presence of a novel object are typically also faster to habituate to the presence of an initially novel object when it is presented with food on multiple consecutive days (Kimball et al., 2022; Krajcir et al., 2024). This suggests that, at least in some species, initial neophobia and neophobia attenuation assays can both reflect an overall 'neophobic' versus 'non-neophobic' phenotype that likely affects many different aspects of neurobiology and behavior.

Conclusions and future directions

Although a substantial body of research has investigated potential mechanisms underlying neophobia, most of this work has focused on laboratory rodents in the context of taste and odor neophobia. Some studies have also been done in birds using object, food and spatial neophobia paradigms, especially on glucocorticoids and neural substrates of neophobia. However, very little research thus far has examined mechanisms of neophobia in non-rodent mammals, amphibians, reptiles or fish, even though variation in neophobia exists within and among these taxa (Candler and Bernal, 2015; Roy et al., 2017; Schaffer et al., 2021; Szabo and Ringler, 2023). More studies in these groups would provide insight into which mechanisms are evolutionarily conserved and which are species specific. Further, artificial selection for laboratory environments has affected at least some types of neophobia and their neuroendocrine mechanisms in domesticated rodents (Kiyokawa et al., 2024; Koizumi et al., 2019). Therefore, a full understanding of how ecological and evolutionary

pressures shape the mechanisms of neophobia will also require examining more wild and wild-derived species.

The fact that ecological and biomedical studies tend to use different neophobia paradigms and study species complicates the search for common mechanisms. An additional complicating factor is that some authors have considered responses to experimental paradigms such as the open field test, elevated plus maze and marble burying assay to reflect neophobia, whereas others (ourselves included) consider it more likely that these assays measure anxiety, or other 'neophobia-adjacent' behaviors. Several commonly used assays also probably measure neophilia, rather than neophobia (see Table 1). This lack of clarity can produce findings that are interpreted as neophobia related even when they are examining other behavioral processes. Standardized assays that include a motivation for animals to approach novelty, such as access to food, water, offspring or other rewarding stimuli, can help distinguish neophobia from neophilia (Kimball and Lattin, 2024).

Despite these limitations and methodological challenges, some consistent patterns in neural substrates and neuroendocrinology emerge across taxa and studies. The regions of the brain most implicated in neophobia are the amygdala (in birds, the medial ventral arcopallium) and the hippocampus, two regions crucial in fear conditioning and memory consolidation (Antoniadis and McDonald, 2001; Chaaya et al., 2018). Also important are the mammalian insular and gustatory cortices, which are involved in taste perception (de Araujo et al., 2012). The role of these brain regions in neophobia is further determined by the specific brain subregion (e.g. the dorsal versus ventral hippocampus in rodents) and the type of novelty examined (e.g. novel tastes versus novel environments) (Fig. 1).

Serotonergic pathways have also been shown to mediate neophobia in complex ways in both birds and mammals, while dopamine mostly seems to reduce neophobia. In rodents, glutamate appears to be essential for the attenuation of taste neophobia, and noradrenaline also plays a key role, although context (i.e. spatial versus food neophobia) is critical. Similar context dependence is seen in glucocorticoids, which differentially affect responses to novelty in birds based on the type of novelty presented (e.g. object versus food), the species, developmental stage and sex examined, and whether baseline or acute glucocorticoids were assessed or manipulated.

In conclusion, despite increased research on neophobia over the past two decades (Kimball and Lattin, 2024), our understanding of the mechanisms underlying this critical behavior is still incomplete and taxonomically biased. Furthermore, most studies have focused on isolated brain regions, neurotransmitters and hormones, and few researchers have investigated how interactions among these different systems affect neophobia (e.g. interactions among glucocorticoids, noradrenaline, glutamate and dopamine; Mora et al., 2012) or how neophobia mechanisms might be affected by ecological variables such as habitat type, diet and sociality.

Another major theme that emerges from the literature is that not all types of neophobia are affected by the same underlying mechanisms: food, object, taste, odor and spatial neophobia sometimes rely on distinct neural substrates and neurotransmitters (Figs 1 and 3). Initial neophobia responses and neophobia attenuation are also often mediated differently. Future work should therefore broaden the taxonomic scope of mechanistic research on neophobia, further investigate shared and distinct mechanisms for different types of neophobia, and explore how different neuroendocrine mechanisms are affected by environmental and ecological factors.

Competing interests

The authors declare no competing or financial interests.

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ECR Spotlight

This article has an associated ECR Spotlight interview with Marquise Henry.

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